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ODD END SEMESTER EXAMINATION DECEMBER 2025

Name of the Course: **B.Tech.**

Semester: **Third**

Name of the Paper: **Discrete Structures and Combinatorics**

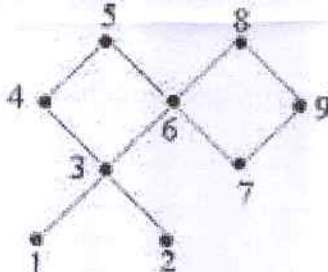
Paper Code: **TMA 316**

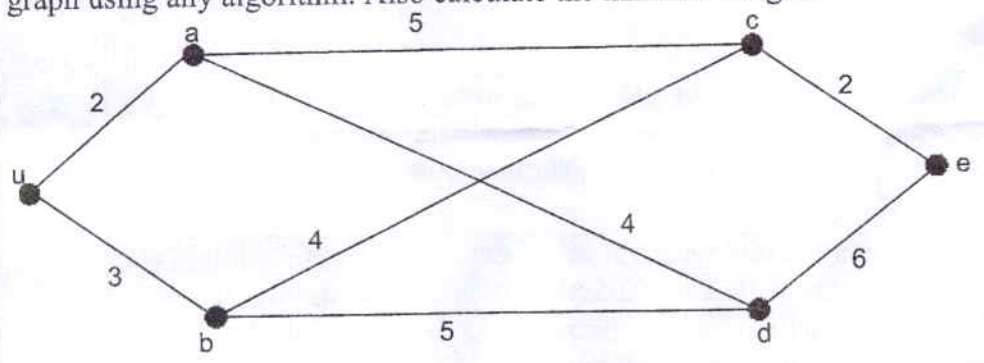
Time: **3 Hours**

Maximum Marks: **100**

Note:

- (i) All questions are compulsory.
- (ii) Answer **any two sub questions** among a, b & c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks.

Q1		(20 marks)	CO1
(a)	If R is an equivalence relation in a set A, then prove that R^{-1} is also an equivalence relation in A.		
(b)	Let $X = \left\{ x : x \in R \text{ and } -\frac{\pi}{2} \leq x \leq \frac{\pi}{2} \right\}$ and $Y = \{ y : y \in R \text{ and } -1 \leq y \leq 1 \}$. Show that mapping $f : X \rightarrow Y$ defined by $f(x) = \sin x$ is bijective mapping.		
(c)	Define Hasse Diagram. Consider the following Hasse diagram Fig. and the set $B = \{3, 4, 6\}$. Find, if they exist,  (i) all upper bounds of B; (ii) all lower bounds of B; (iii) the least upper bound of B; (iv) the greatest lower bound of B;		
Q2		(20 marks)	CO2
(a)	Assuming half the population of a town consumes chocolates and that 100 investigators each take 10 individuals to see whether they are consumers, how many investigators would you expect to report that 3 people or less were consumers?		
(b)	A transport company needs tyres for its trucks and the probabilities are 0.18, 0.22, 0.02, 0.23, 0.30 and 0.05 that it will buy Goodyear tyres, Uniroyal tyres, Michelin tyres, General tyres, Goodrich or Armstrong tyres. Find the probability that it will buy (i) Goodyear or General Tyres. (ii) Uniroyal, Michelin or Goodrich tyres.		

	(iii) Goodyear, Uniroyal, Armstrong or Goodrich tyres.	
(c)	In a certain factory turning out razor blades, there is a small chances of 0.002 for any blade to be defective. The blades are supplied in packets of 10. calculate the approximate number of packets containing no defective, one defective and two defective blades in a consignment of 10,000 packets.	
Q3	(20 marks)	
(a)	(i) Prove that $(p \Leftrightarrow q) \Leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p)$ is a tautology. (ii) Show that $(p \rightarrow r) \wedge (q \rightarrow r)$ and $(p \vee q) \rightarrow r$ are logically equivalent.	CO3
(b)	Prove by mathematical induction $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$.	
(c)	Check the validity of the following argument: "If 8 is even then 2 does not divide 9. Either 7 is not prime or 2 divides 9. But 7 is prime, therefore 8 is odd."	
Q4	(20 marks)	
(a)	Show that the set of all positive rational numbers forms an abelian group under the composition defined by $a * b = \frac{ab}{4}, a, b \in Q^+$	CO4
(b)	(i) State and Prove Lagrange's theorem. (ii) Find the order of each element of the multiplicative group $\{G=1, \omega, \omega^2\}$	
(c)	Solve the recurrence relation by using Generating function. $a_r - 4a_{r-1} + 4a_{r-2} = (r+1)^2, r \geq 2$	
Q5	b(20 marks)	
(a)	Define with examples i. Pseudo graph ii. Binary Tree iii. Trail iv. Complete Graph v. Bipartite Graph	CO5, CO4
(b)	What is a weighted graph? Find the minimal spanning tree for the given graph using any algorithm. Also calculate the minimal weight. 	
(c)	(i) Prove that, the sum of degrees of all vertices in a graph G is twice the number of edges in G. (ii) Prove that every Cyclic group is an abelian group.	